

Nisan Stiennon

Mathematician • Research Engineer

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EXPERIENCE

- 2021 – present **Independent Researcher** • Mathematics & cooperative AI half-time
Building a language for open-source games and agents, with a denotational semantics and results on the conditions for cooperation. Joint work with Scott Garrabrant.
- Vitalik Buterin Postdoctoral Fellowship in AI Existential Safety, 2022–2025.
 - Center on Long-Term Risk: research grant, 2021–2022; Summer Research Fellow, 2022.
- 2022 – 2026 **ML Engineer** • Encultured AI Berkeley, half-time
- Game development in Rust.
 - Collaborative AI fiction app.
 - LLMs for equities trading.
 - LLMs for medical research & patient advocacy.
 - LLM aggregator: themultiplicity.ai
- 2018 – 2020 **Member of Technical Staff** • OpenAI San Francisco
- RLHF for LLMs.
- 2014 – 2018 **Software Engineer, L4** • Google San Bruno
- ML engineering for YouTube personalized notifications.

EDUCATION

- 2007 – 2013 **Ph.D. in Mathematics** • Stanford University Palo Alto
Thesis: *The moduli space of real curves and a $\mathbb{Z}/2$ -equivariant Madsen-Weiss theorem*. Advised by Prof. Søren Galatius.
- 2003 – 2007 **B.S. in Mathematics & Physics** • University of Michigan Ann Arbor

SELECTED PUBLICATIONS

- 2021 J. Wu, L. Ouyang, D. M. Ziegler, **N. Stiennon**, R. Lowe, J. Leike, and P. Christiano. *Recursively Summarizing Books with Human Feedback*. arXiv: 2109.10862.
- 2020 **N. Stiennon**, L. Ouyang, J. Wu, D. Ziegler, R. Lowe, C. Voss, A. Radford, D. Amodei, and P. F. Christiano. *Learning to summarize with human feedback*. In: *NeurIPS*. arXiv: 2009.01325.
- 2019 D. M. Ziegler, **N. Stiennon**, J. Wu, T. B. Brown, A. Radford, D. Amodei, P. Christiano, and G. Irving. *Fine-Tuning Language Models from Human Preferences*. arXiv: 1909.08593.

SKILLS

- Languages** Python, Rust, C++.
- ML/AI** LLM fine-tuning, RLHF, agent scaffolding, PyTorch.
- Mathematics** Domain theory, category theory, algebraic topology, probability, functional analysis.